

C5 - MULTIMODAL RECOGNITION

Image Captioning and Diffusion Models

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MASTER IN
COMPUTER VISION
Barcelona



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Motivation

Image Captioning

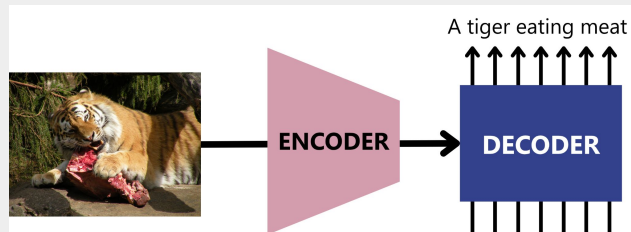


A person riding a motorcycle on a dirt road.

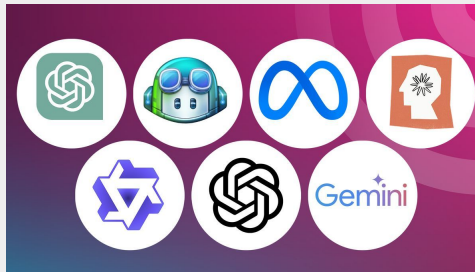
- Translating pixels into natural language (Bridging **CV** & **NLP**)
- Crucial for Accessibility (screen readers), Image Retrieval, Robotics...

How We Tackle It

- Encoder-Decoder Frameworks



- Multimodal LLMs



Dataset: VizWiz

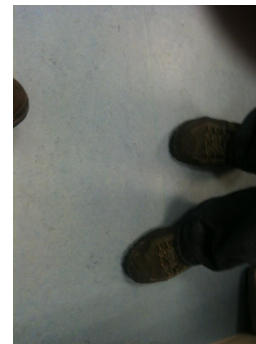
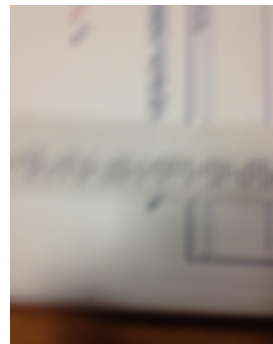
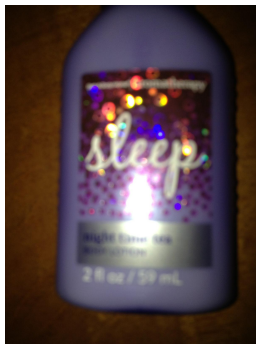
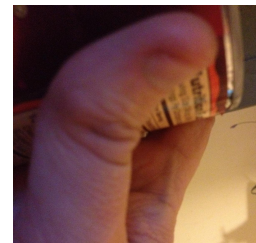
Definition

VizWiz is a VQA dataset built from photos taken by **blind and low-vision people** using their smartphones.

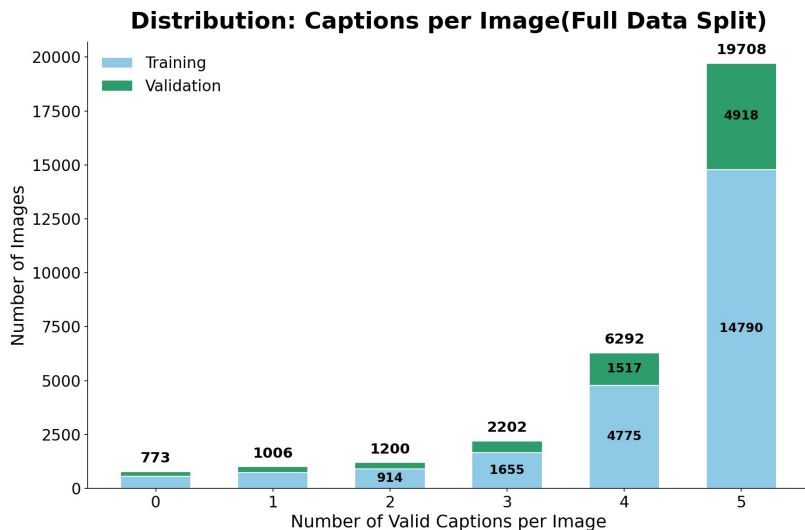
Why it matters

VizWiz reflects **authentic degradation**:

- Extreme Blur
- Poor Lighting
- Fingers on lens
- Off-frame subjects
- Flash Glare
- Unanswerable



Dataset: VizWiz



TRAINING SET	
23431	117155
IMAGES	CAPTIONS

TEST SET	
7542	33186
IMAGES	CAPTIONS
Official validation set used as test split	

Key Observations

- Images with **no caption** are removed.
- The official **test set** is **not used**.
- **Imbalance** in the number of **captions per image**, which may affect models.

Metrics

BLEU

Bilingual Evaluation Understudy

Measures n-gram precision between generated and reference captions.

ROUGE-L

Recall-Oriented Understudy for Gisting Evaluation

Measures longest common subsequence for recall-oriented evaluation.

CIDEr

Consensus-based Image Description Evaluation

Uses TF-IDF to weight n-grams, rewarding distinctive descriptions.

METEOR

Selected

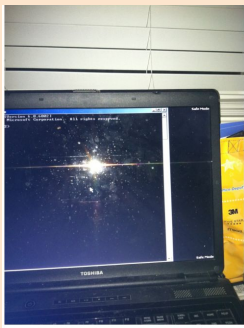
Metric for Evaluation of Translation with Explicit ORdering

Considers synonyms, stemming and word order beyond exact matches.

Training Strategy

Training

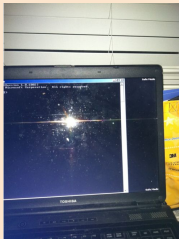
GT Captions



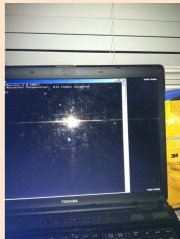
Here is a picture of a laptop screen with safe mode.

A Toshiba laptop in safe mode with command prompt open.

Replicated Images



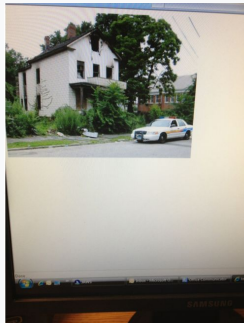
Here is a picture of a laptop screen with safe mode.



A Toshiba laptop in safe mode with command prompt open.

Testing

GT Captions



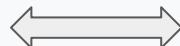
*A house with some police car in the monitor window.
computer display monitor displaying old home and police vehicle.*

Model Prediction

A police car is parked in front of an abandoned house with overgrown vegetation.

Metrics Calculation

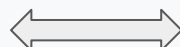
A house with some police car in the monitor window.



A police car is parked in front of an abandoned house with overgrown vegetation.

METEOR:
0.5409

computer display monitor displaying old home and police vehicle.



A police car is parked in front of an abandoned house with overgrown vegetation.

METEOR:
0.0526



Motivation

Goals

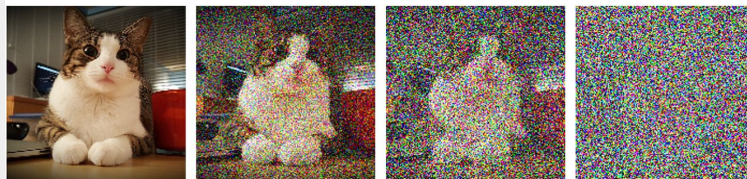
- **Problem:** VizWiz dataset is irregular and noisy.
- **Solution:** Generate **targeted synthetic data** to patch specific flaws in our captioning models.



*An unedited, amateur smartphone snapshot of a **white bread maker sits on a black counter**. Casual, everyday indoor lighting, slightly off-center framing, messy background.*

Diffusion

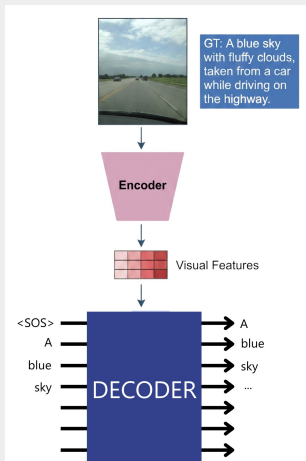
- Generates high-fidelity **images from text**.
- Bridges **NLP** (text analysis) and **generative models** (image synthesis) to produce context-aware visuals.



Methodology & Approach: A Four-Stage Evolution

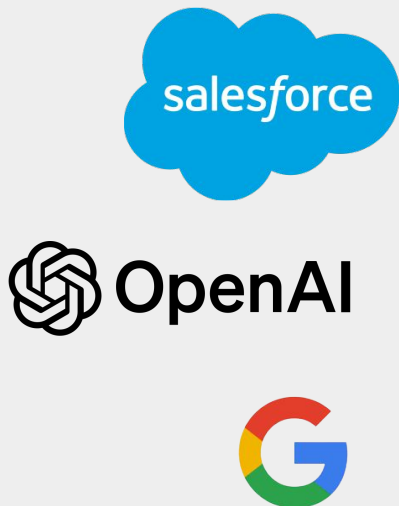
1- Basic Frameworks

- Basic encoder-decoder architecture
- Train from scratch
- Bridge modality gap



2- Pre-Trained VLMs

- Domain adaptation
- Fine-tune for noisy data



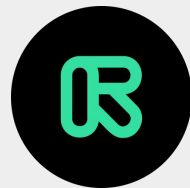
3- Multimodal LLMs

- Scale reasoning
- Zero-shot vs. PEFT (LoRA)



4- Diffusion Augmentation

- Target synthetic generation
- Fix dataset flaws



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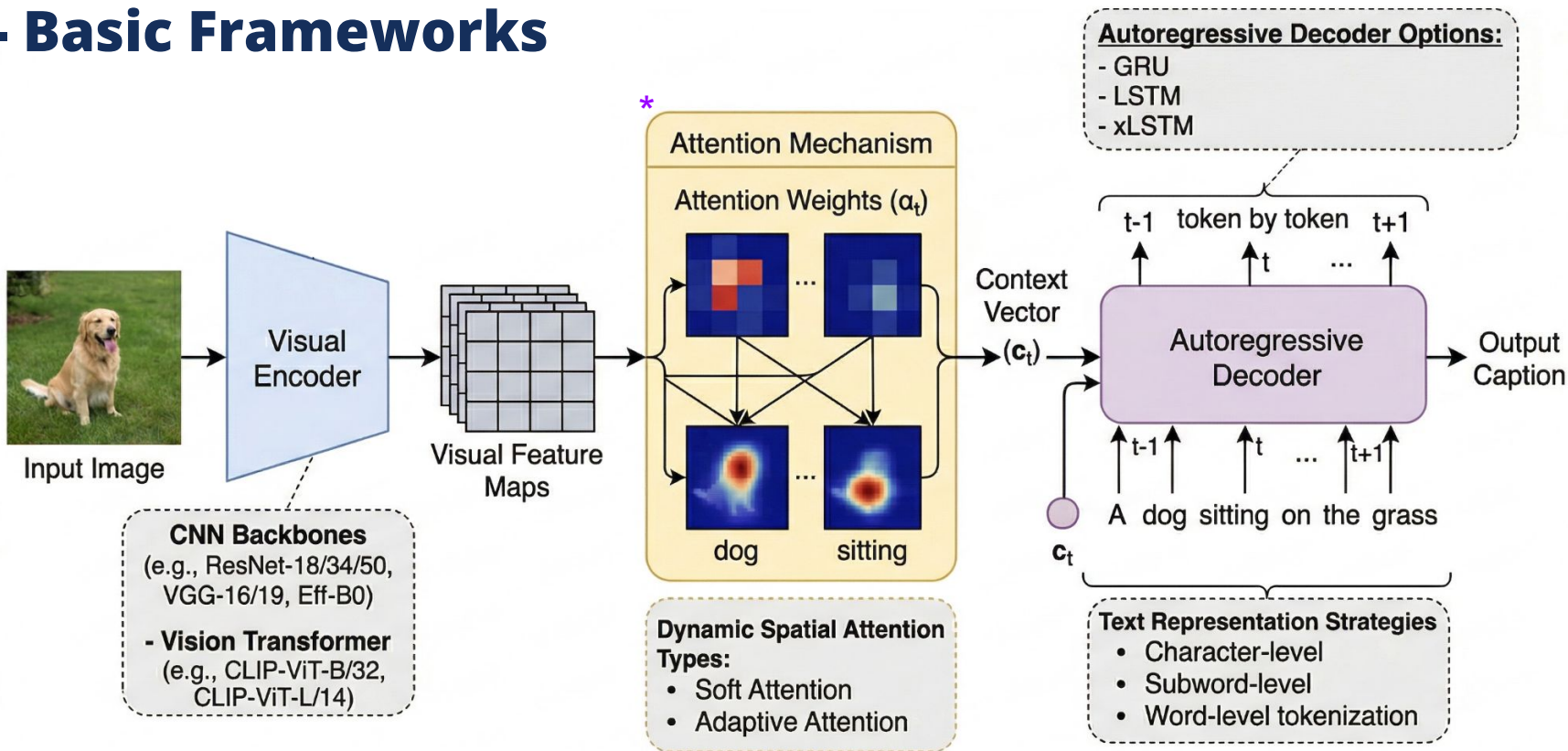
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Conclusions

1- Basic Frameworks



* K. Xu, J. Ba, R. Kiros, K. Cho, A. Courville, R. Salakhutdinov, R. Zemel, and Y. Bengio, "Show, attend and tell: Neural image caption generation with visual attention," in *International Conference on Machine Learning (ICML)*, 2015, pp. 2048-2057.

* J. Lu, C. Xiong, D. Parikh, and R. Socher, "Knowing when to look: Adaptive attention via a visual sentinel for image captioning," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 3242-3250.

1- Basic Frameworks: Quantitative Results

0) Baseline (ResNet18 + GRU + Characters)

- Optimizer: Adam
- LR: 5e-5
- Scheduler: Plateau
- Batch size: 128

1) Encoder (CLIP-ViT-b32)

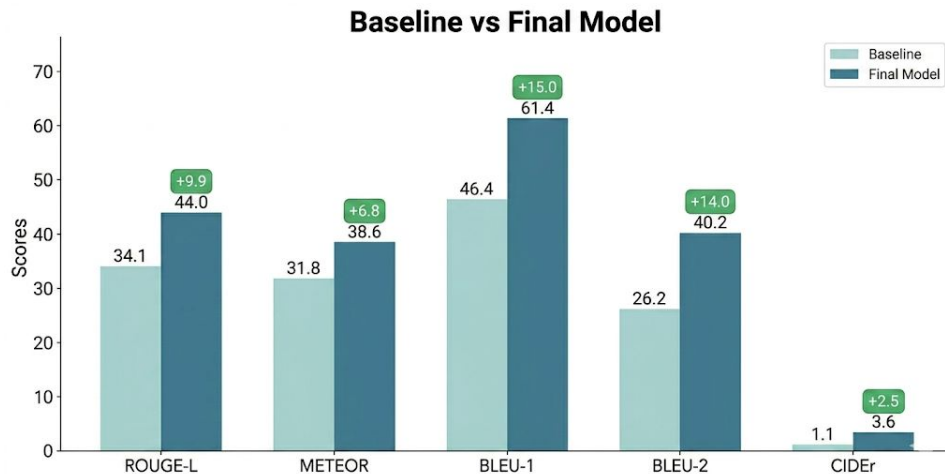
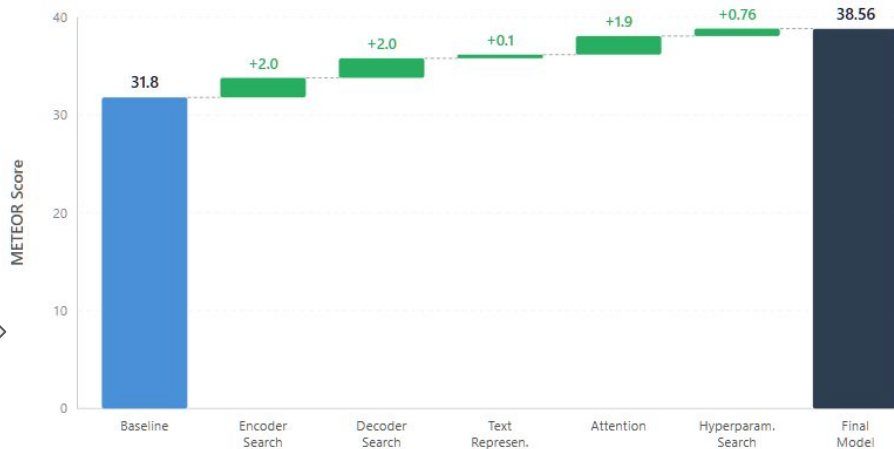
2) Decoder (LSTM: 3 layers, 1024 dim)

3) Text Representation (Word)

4) Attention (Adaptive Attention)

5) Hyperparameter Search:

- Optimizer: Adam
- Lr: 1e-4
- Scheduler: Cosine
- Decoder Strategy: Beam Search



1- Basic Frameworks: Qualitative Results

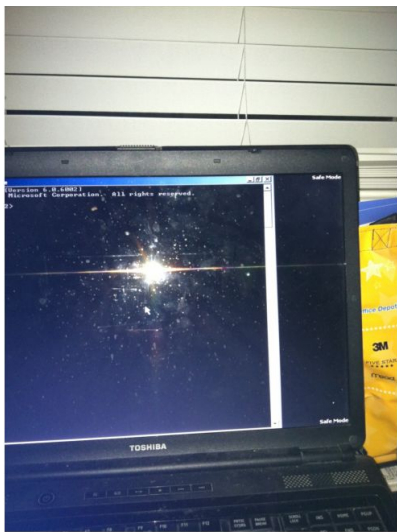


Caption - Baseline

A pair of black shoe with a blue screen and a white shirt and a white shirt and a white sheet.

Caption - Final

a picture of a man in a red dress standing outside .

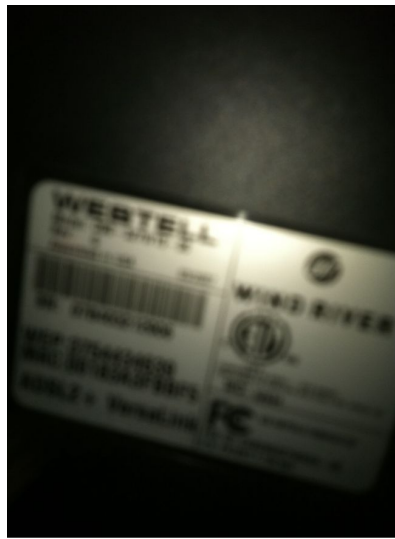


Caption - Baseline

A black computer monitor with a black and white striped shirt and a black background

Caption - Final

a computer screen with a blue background and white text is shown .

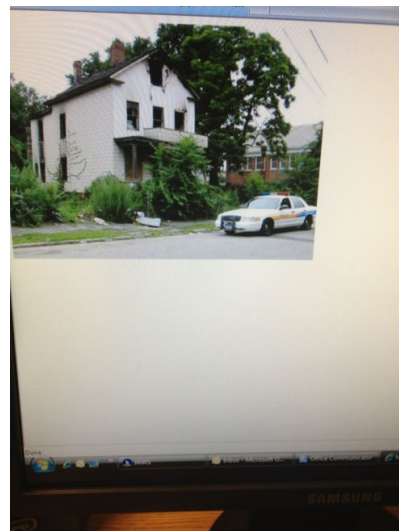


Caption - Baseline

A package of food is on top of a table.

Caption - Final

a label with a barcode and a barcode on a black surface



Caption - Baseline

A computer screen with a black and white striped shirt and a black background.

Caption - Final

a computer screen with a picture of a house and trees

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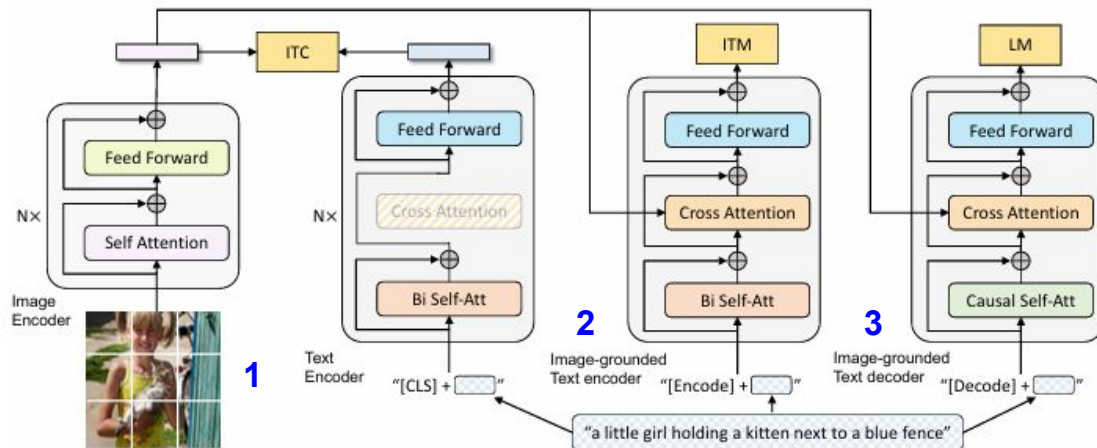
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2- Pre-Trained VLMs



Shift to Transformers

Overcoming CNN-RNN limitations (long-range dependencies, data hunger).

Models Evaluated

- ViT-B/16 + GPT-2
- ViT-B/16 + BERT
- BLIP

Adaptation Strategies

- Zero-shot inference
- Targeted Fine-Tuning configurations

BLIP*

Multimodal Mixture of Encoder-Decoder (MED)

1. Unimodel encoder (image/text)
2. Image-grounded text encoder
3. Image-grounded text decoder

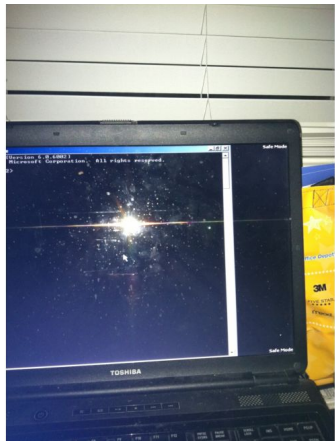
* J. Li, D. Li, C. Xiong, and S. Hoi, "BLIP: Bootstrapping Language-Image Pre-training for Unified Vision-Language Understanding and Generation," in Proceedings of the 39th International Conference on Machine Learning (ICML), 2022, pp. 12888–12900.

2- Pre-Trained VLMs: Quantitative results

Model	BLEU-1	BLEU-2	ROUGE-L	METEOR	#params	Encoder Finetuned	Decoder Finetuned	Improvement vs Best Model S1
Best w/ Attention: CLIP-ViT-b32- LSTM-Word- Adaptive Attention	61.4	40.2	44.0	38.6	104 M	✗	✓	0.0
ViT-B/16 - GPT2	51.8	31.9	37.1	27.4	210 M	✗	✗	-11.2
ViT-B/16 - BERT	39.8	17.0	27.1	22.5	196 M	✗	✗	-16.1
BLIP	55.2	37.4	42.5	32.3	247.1 M	✗	✗	-6.3
BLIP Finetuned	61.4	40.2	44.0	39.8	247.1 M	✓	✓	1.2

 : Best Model Stage 1
  : Current Week Models

2- Pre-Trained VLMs: Qualitative results



Caption - **Best S1- BAD**

a laptop screen is on top of a laptop

Caption - **BLIP PreTr - BAD**

a laptop computer sitting on a desk with a screen showing a meteor

Caption - **BLIP FT - BETTER**

a laptop computer with a blue screen and a yellow box in the background

Caption - **Best S1- GOOD**

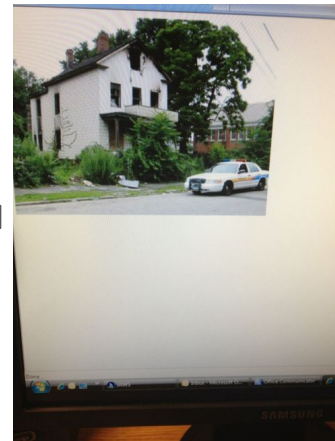
a computer screen with a picture of a car on it

Caption - **BLIP PreTr - GOOD**

a computer screen with a picture of a house.

Caption - **BLIP FT - BETTER**

a computer screen showing a picture of a house and a car



GT Captions

- here is a picture of a laptop screen with safe mode.
- a Toshiba laptop in safe mode with command prompt open.

Key Takeaways

- **Reduced Hallucinations:** Pretrained models struggle with noise.
- **Higher Specificity:** Fine-tuning captures more valid details.
- **Limitation:** Missing text/brand context.

GT Captions

- A house with some police car in the monitor window
- computer display monitor displaying old home and police vehicle.



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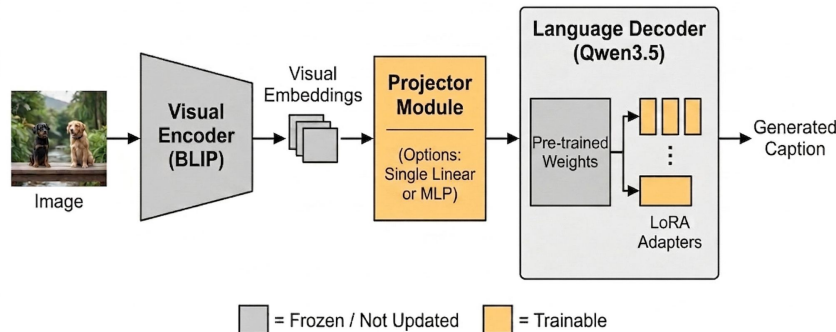
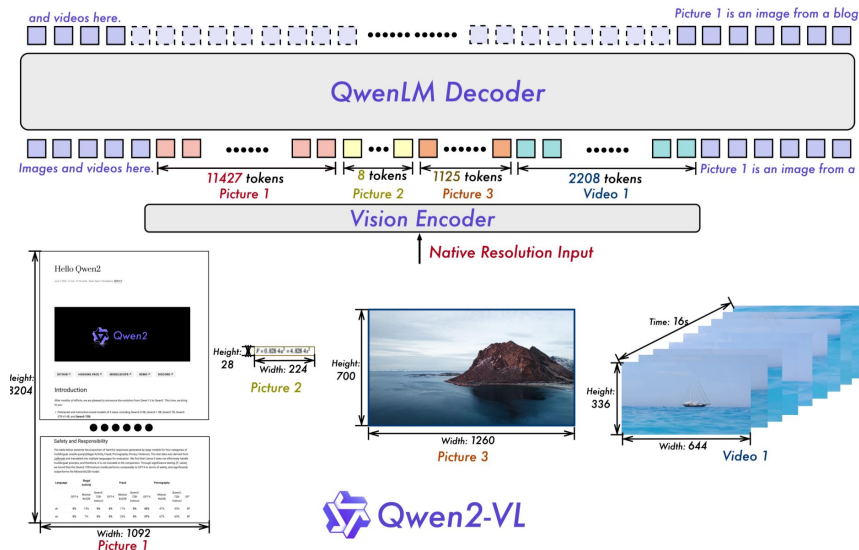
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3- Multimodal LLMs



Fine-tuning

LoRA Adapters: Efficient parameter updates for the LLM.

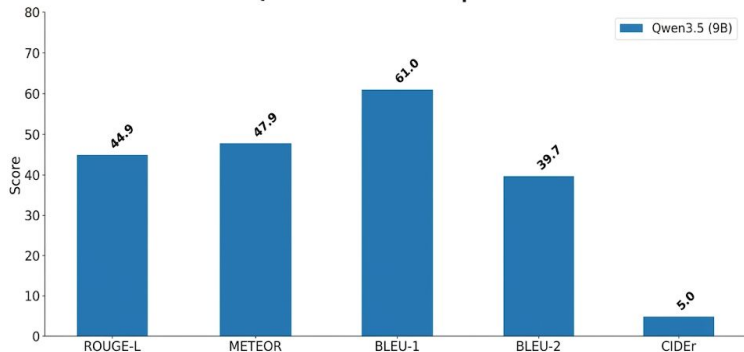
- **BLIP Encoder:** Utilized our fine-tuned BLIP model from the previous week.
- **Qwen 3.5 Decoder:** The generative component, we tried 0.8B, 2B and 4B models.

Pre-trained Qwen Models

We explored various **pre-trained Qwen models (2, 2.5, 3 & 3.5)** as the base LLM decoder. These models provide the foundational reasoning and linguistic capabilities for the multimodal system.

3- Multimodal LLMs: Quantitative Results

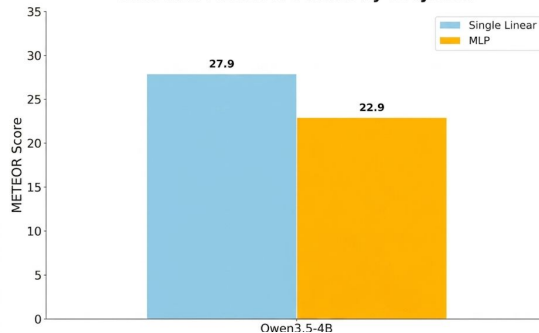
Qwen VL Model Comparison



Pretrained Models

Very good results, but massive to train from scratch or fine-tune with limited resources.

Task 2.2: METEOR Scores by Projector

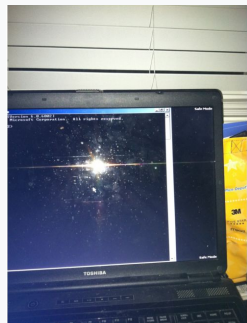


Fine-tuned Models & Modality Collapse

- BLIP vision encoder is kept frozen.
- Training limited to multimodal projector and LoRA adapters.
- Modality collapse: Model treats projector outputs as random noise, ignoring image tokens.

Showing best performing model results in terms of METEOR for both cases.

3- Multimodal LLMs: Qualitative Results



GT Captions

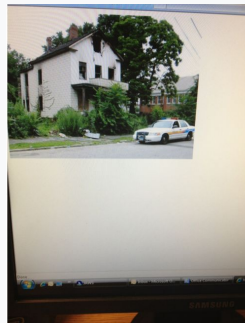
- Here is a picture of a laptop screen with safe mode
- A Toshiba laptop in safe mode with command prompt open.

Qwen3.5 (Pre-trained 9B) ✓

A Toshiba laptop displaying a Microsoft command prompt with "Safe Mode" visible on the screen.

Qwen3.5 (Fine-tuned 4B)

The image displays a close-up of a dark glass bottle, likely containing wine or beer, set against a blurred background.



GT Captions

- A house with some police car in the monitor window
- computer display monitor displaying old home and police vehicle.

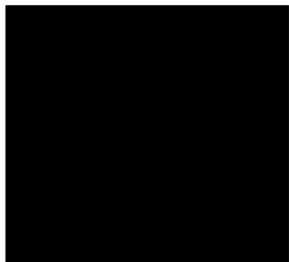
Qwen3.5 (Pre-trained 9B) ✗

A police car is parked in front of an abandoned house with overgrown vegetation.

Qwen3.5 (Fine-tuned 4B)

The image displays a close-up of a dark, textured fabric, likely a fleece or wool blanket, with a subtle pattern of small dots.

Ref: (Blank/Black image - No Reference)



Caption - Qwen3.5

"The image displays a close-up view of a person's face, specifically focusing on the right side of the face. The lighting is bright, casting distinct..."



Modality Collapse



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4- Diffusion Augmentation

SD 1.5

Architecture

Standard U-Net (~860M parameters).

General

Baseline model for initial testing at 512x512 resolution.

Role

Proof-of-concept for augmentation.

SD XL

Architecture

Dual-encoder ensemble (~3B parameters).

General

Jump in fidelity & prompt adherence (1024px).

Drawback

Higher cost and slower generation.

★ SDXL Turbo

Real-time Performance

High quality in just 1–4 steps (ADD Distillation).

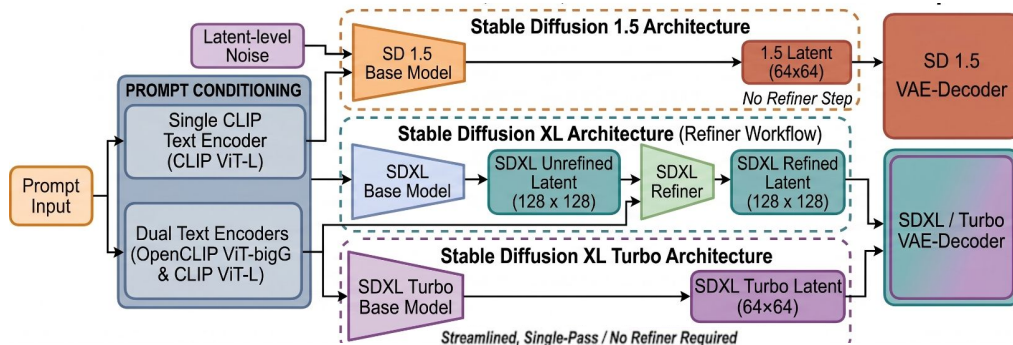
Optimal Balance

Fidelity of SD XL with extreme speed.

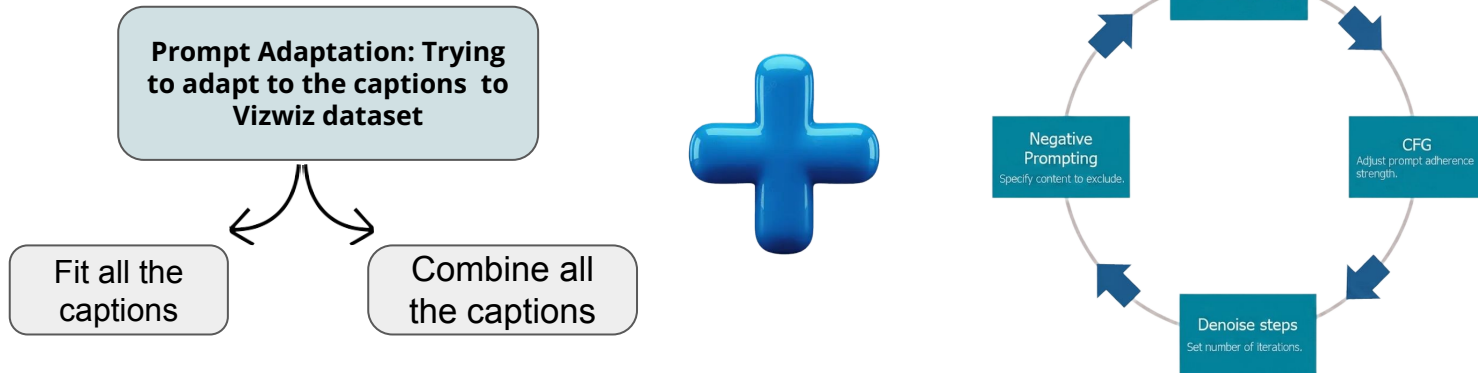
Selection

Rapidly generate volumes without bottlenecks.

Revised SD 1.5, SDXL, and SDXL Turbo Architecture Pipelines



4- Diffusion Augmentation



Model	Generating time/image
SD v1.5	34s
SDXL 1.0 base	151s
SDXL Turbo	1.2s

SD 1.5



SD XL



SD XL Turbo

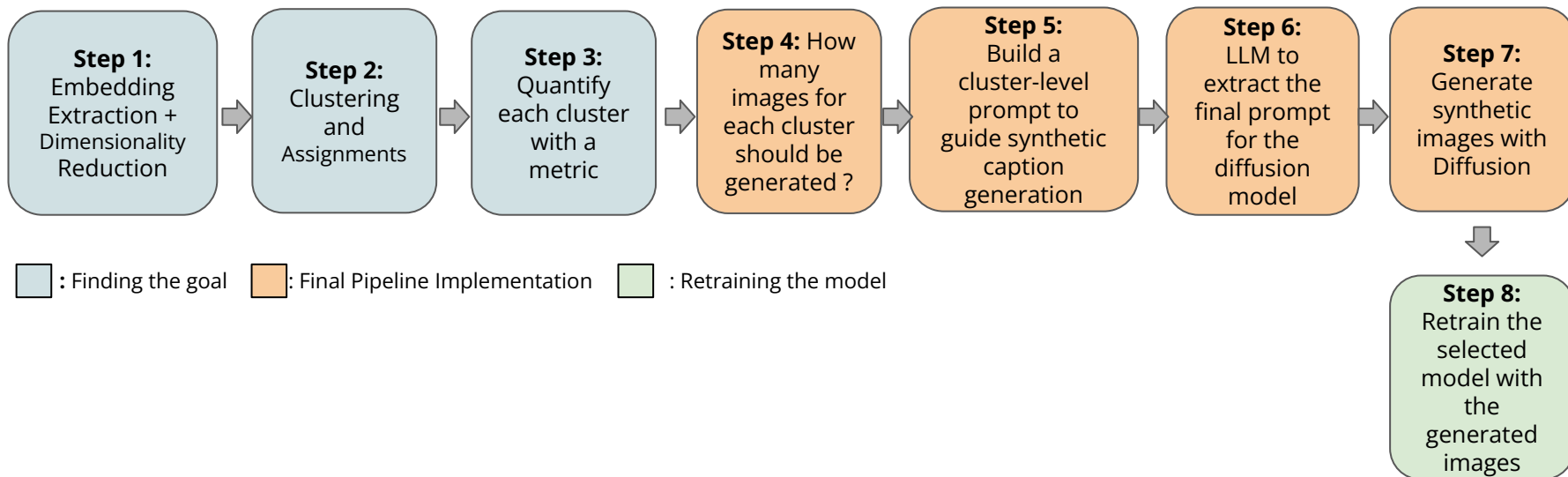


4- Diffusion Augmentation

Selected Model

We selected **fully fine-tuned BLIP model**

Pipeline

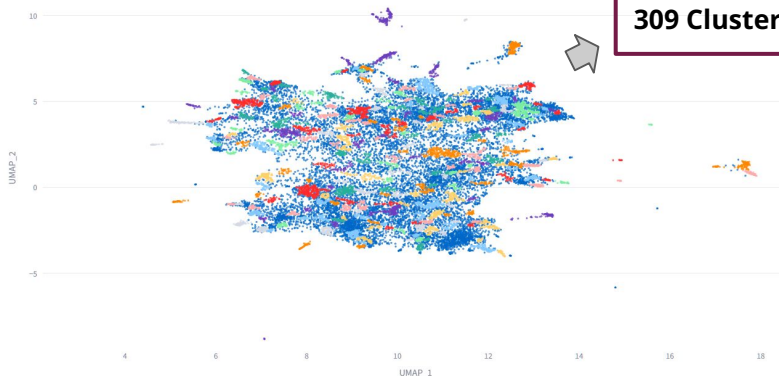


4- Diffusion Augmentation - Step 1 & 2

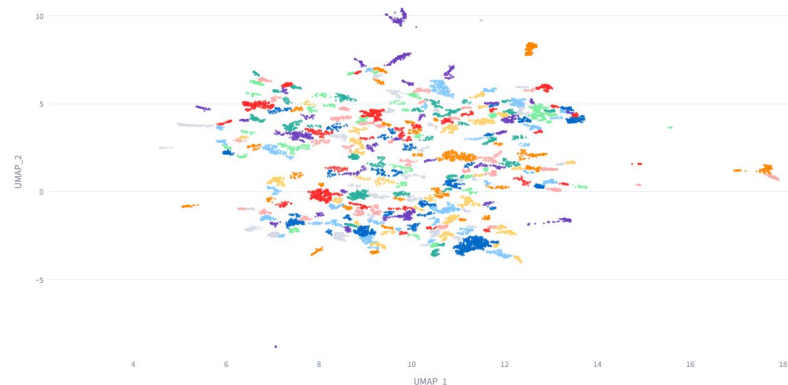
Step 1 & 2

- 1) Extract embeddings with a **pretrained CLIP** (in **training** and **validation**)
- 2) Fit a **UMAP** in the **training** to reduce dimensionality to **2D**
- 3) Use the **fitted UMAP** to project **validation data** into the **same embedding space**
- 4) Fit a clustering algorithm (**HDBSCAN (*)**) on the **training data**
- 5) Assign **validation data** into the **corresponding clusters**
- 6) **Remove cluster -1** corresponding to **61.2% of the data**

2D Training Samples (with Noise Cluster (-1)) (*)



2D Training Samples (without Noise Cluster (-1)) (*)

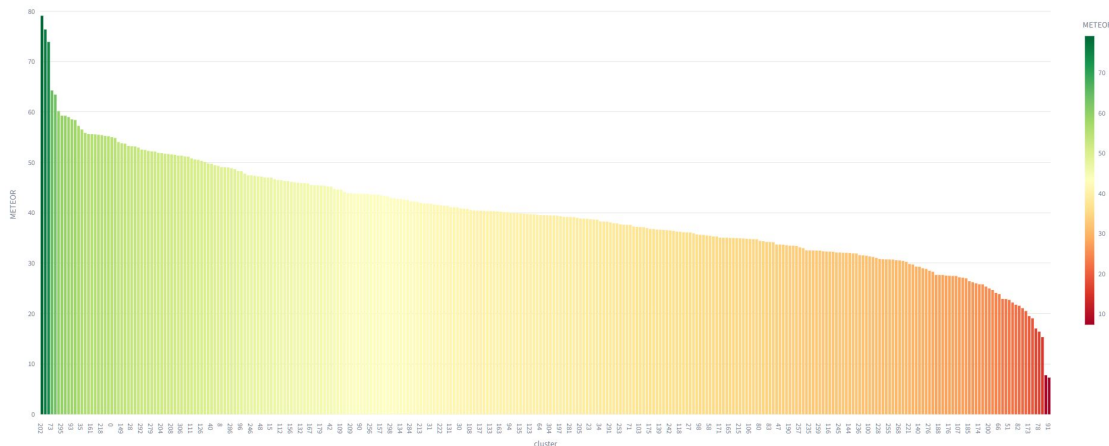


4- Diffusion Augmentation - Step 3

Step 3

- 1) Run **BLIP** on the validation split.
- 2) Compute the **average METEOR score** for the validation samples assigned to **each cluster**.

Meteor Score by cluster (sorted) (*)



Goal: Boost overall METEOR and reduce inter-cluster performance gaps using targeted synthetic training data.

4- Diffusion Augmentation - Step 4 & 5 & 6

Step 4

- 1) Synthetic images are generated only for clusters **below the average METEOR (39.8)**.
- 2) For the remaining clusters, the number of samples is **determined by the formula shown on the right**.



$$\text{inverted_score} = 1 - \frac{\text{METEOR}_i - \text{METEOR}_{\min}}{\text{METEOR}_{\text{threshold}} - \text{METEOR}_{\min}}$$
$$n_{\text{samples}} = \text{round}(\text{min_samples} + \text{inverted_score} \times (\text{max_samples} - \text{min_samples}))$$

For each cluster selected to generate images:

Step 5 & 6

- 1) Generate **template prompts** for each cluster (see right) (**step 5**).
- 2) Feed these templates into an LLM (**Qwen 3.5 9B**) to produce the **final captions used as input for the diffusion models (step 6)**



Generate **{N}** diverse image captions with maximum 17 words each. Integrate **{K}** of the following visual characteristics/qualities naturally into only 3 of the generated captions: **{quality_1, quality_2, ..., quality_K}**.

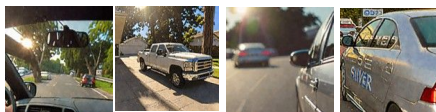
Here you have examples of correct captions belonging to this cluster to guide the semantic content of your new captions:

- **{caption_1}**
- **{caption_2}**
- **...**
- **{caption_M}**

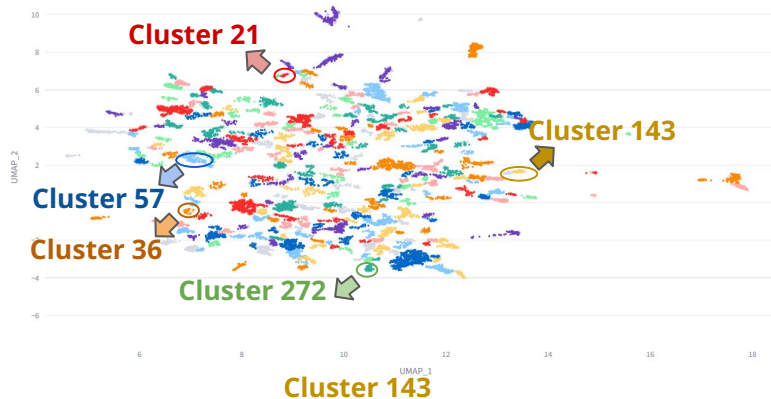
4- Diffusion Augmentation - Step 7 A

- * Training samples
- * 2S0CFG Diffusion Model
- * 4S1CFG Diffusion Model

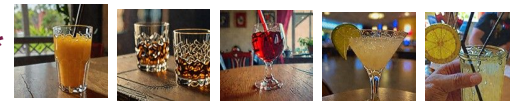
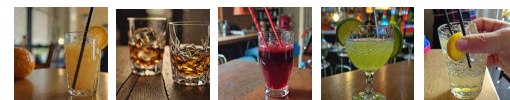
Cluster 21



2D Training Samples (without Noise Cluster (-1))



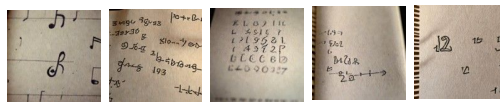
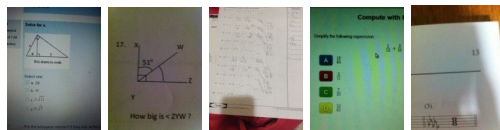
Cluster 36



Cluster 57



Cluster 143



Cluster 272



4- Diffusion Augmentation - Step 7 B

Evaluation

- 1) Extract embeddings with a **pretrained CLIP**
- 2) Use the **fitted UMAP** to project **generated images** into the **same embedding space**
- 3) Assign **generated images** into the **corresponding clusters (using euclidean distance to the centroid)**
- 4) Compute metrics

Results

Model ID	Top 1	Top 5	Top 10	Top 15	Top 20
2S0CFG	10.3	25.8	35.4	41.2	45.9
4S1CFG	10.2	24.9	35.5	40.3	44.4

4- Diffusion Augmentation - Step 8

Training Strategies

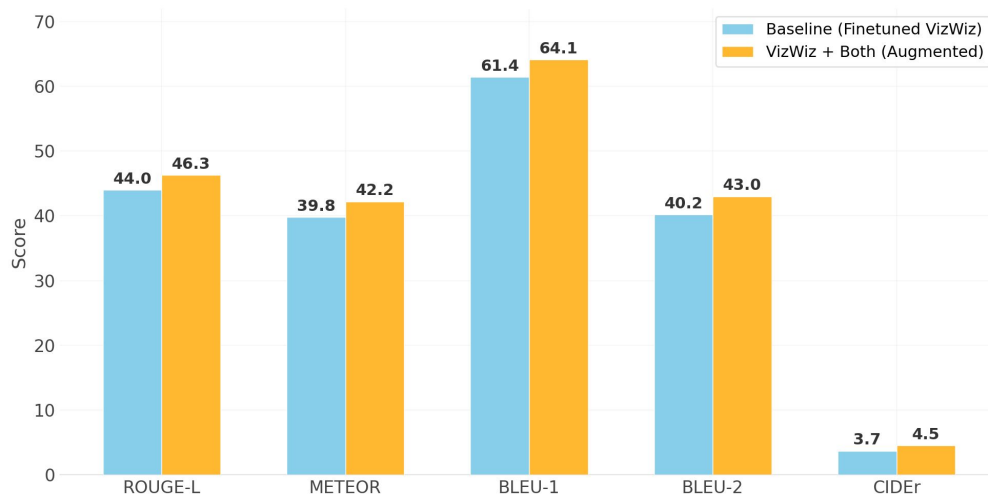
1. Data Augmentation (Mixed):

- VizWiz + 2S0CFG
- VizWiz + 4S1CFG
- **VizWiz + Combined Synthetic Data (Both)**

2. Fine-Tuning on Synthetic (Isolated):

- 2S0CFG Only
- 4S1CFG Only
- Combined Synthetic Data Only (Both)

BLIP Finetuned Models Comparison

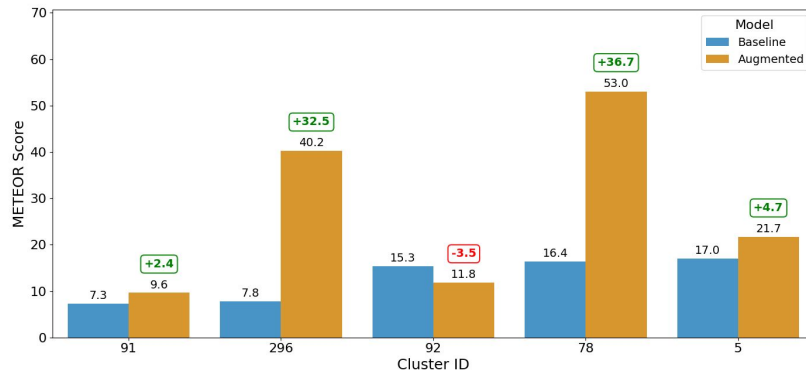


4- Diffusion Augmentation - Step 8

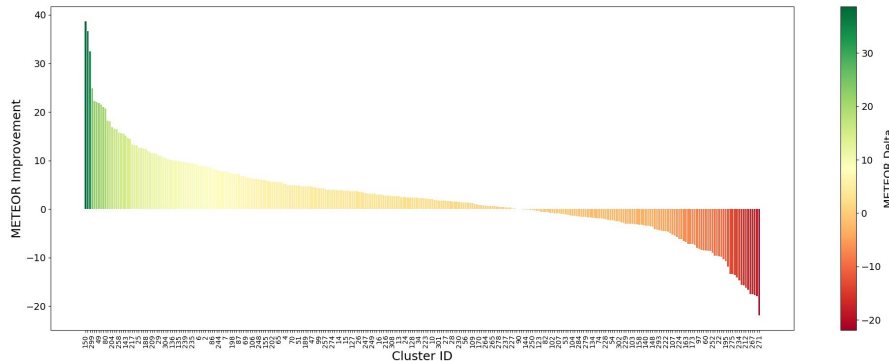
Cluster Improvements

- **4 out of 5** worst clusters **improved significantly** (up to **+36.7** METEOR)
- **~65%** of all clusters showed **positive gains**
- Minor regressions are **expected due to generative noise and domain shift**

Top 5 Worst Performing Clusters: Baseline vs Augmented



METEOR Delta per Cluster (Augmented - Baseline)



4- Diffusion Augmentation - Step 8

Cluster 92



Caption - Baseline

a bag of food is on top of a table.

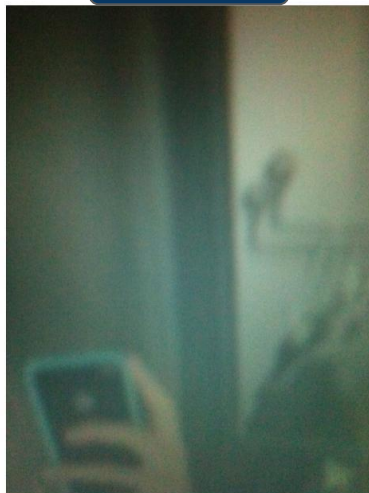
METEOR: 15.75

Caption - Augmented

a bag of food is on top of a table.

METEOR: 15.75

Cluster 78



Caption - Baseline

a close up of a computer screen
with a white and green image.

METEOR: 16.39

Caption - Augmented

a person is holding a cell phone
in their hand.

METEOR: 53.05

Cluster 296



Caption - Baseline

a box of darigo brand darigo
brand darigo brand darigo brand
darigo brand darigo brand

METEOR: 7.75

Caption - Augmented

a package of unidentifiable product with a white lid.

METEOR: 40.24



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Conclusions

- **Full BLIP Fine-Tuning: Essential** to overcome VizWiz's severe **domain shift**.
- **LLMs Excel Zero-Shot**: Qwen3.5 achieved the **highest METEOR** by maximizing **semantic recall**.
- **Modality Collapse**: Linear projectors fail; the model **ignores images** and **hallucinates**.
- **Targeted Data Augmentation is Effective**
- **High Impact from Minimal Data**: This gain was driven entirely by fixing the bottom **27.3%** of the data.

Future Work

- **Joint Training**: Train the encoder, projector, and LLM **end-to-end** (unfreeze vision).
- **Cross-Attention**: Replace linear projectors to better **align embedding spaces**.
- **Density-Aware Sampling**: Instead of purely using average METEOR, factor in the **number of samples per cluster**.
- **Clustering Optimization**: Tune the clustering hyperparameters (e.g., HDBSCAN)
- **Alternative Diffusion Applications**



: Image captioning



: Diffusion augmentation

C5 - MULTIMODAL RECOGNITION

Image Captioning and Diffusion Models

Team 1:

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MASTER IN
COMPUTER VISION
Barcelona





VizWiz: Where state-of-the-art models go to cry. **Valentin**: Goat of AI generated images
Valentin: He is the best diffusion model

